

Analysis of Meteorological Observations from the NISER Automatic Weather Station, Jatni (2025)

EPS101 — Earth and Planetary Sciences | Project #26

Partha Sarathi Mohanta

National Institute of Science Education and Research (NISER),
Jatni, Bhubaneswar, Odisha 752050, India

Instructor: Dr. Jayesh Mahendra Goyal

Abstract. This report presents an eight-question analysis of 15-minute resolution meteorological data recorded at the NISER Jatni Automatic Weather Station (AWS) for the period 1 March – 31 October 2025. Variables examined include surface air temperature, relative humidity, atmospheric pressure, wind speed, wind direction, solar radiation and rainfall. The study characterises pre-monsoon heating, monsoon onset and retreat, and post-monsoon stabilisation through daily and monthly statistics, diurnal cycles and derived quantities (specific humidity, upper-level pressure). No cyclone-class signature was detected; pressure anomalies in late May and late July are attributed to typical monsoon low-pressure systems. All analyses were performed in Python using `pandas`, `NumPy` and `Matplotlib`.

Keywords: monsoon meteorology; automatic weather station; diurnal cycle; specific humidity; barometric formula; Bhubaneswar; Odisha.

1. Table of contents

1	Table of contents	1
2	Introduction	2
3	Data and Methods	2
3.1	Observational Dataset	2
3.2	Processing and Analysis	2
4	Results	3
4.1	Daily Time-Series (Question 1)	3
4.2	Global Extremes (Question 2)	6
4.3	Monthly Extremes (Question 3)	7
4.4	Month of Maximum Rainfall (Question 4)	10
4.5	Specific Humidity: Rainy vs. Winter Month (Question 5)	11
4.6	Diurnal Variation — August 2025 (Question 6)	12
4.7	Cyclone Signature Search (Question 7)	15
4.8	Pressure at 10 km Altitude (Question 8)	16
5	Discussion	17
6	Conclusions	17
7	Acknowledgements	18
		18

2. Introduction

The NISER Jatni AWS records continuous observations at 15-minute intervals using a Campbell Scientific datalogger (station ID `NISER_Jatni`, logger serial 19192). The station is situated on the NISER campus ($\approx 20.1^\circ$ N, 85.7° E, ~ 45 m a.s.l.), located approximately 25 km south-west of Bhubaneswar, Odisha — a region strongly influenced by the Bay of Bengal branch of the Indian Summer Monsoon.

This project analyses data spanning March through October 2025, a window that captures the full arc of the monsoon season: pre-monsoon heating (March–May), onset and active monsoon (June–September), and the withdrawal phase (October). The primary objectives are to (i) identify seasonal and diurnal variability in all recorded parameters, (ii) extract global and monthly extremes, (iii) compute specific humidity and estimate upper-tropospheric pressure, and (iv) assess whether any synoptic-scale disturbance resembling a tropical cyclone is present in the observational record.

3. Data and Methods

3.1 Observational Dataset

The primary dataset is `NISER_Jatni_2025_Min_15.csv`, a comma-separated file with a four-row Campbell Scientific header. Usable records begin on 1 January 2025; this study filters to the window 1 March – 31 October 2025. Table 1 summarises the seven recorded variables and their units.

Table 1: Variables recorded by the NISER Jatni AWS and their units.

Column name	Physical variable	Unit	Aggregation
<code>WS_ms_Avg</code>	Wind speed	m s^{-1}	15-min mean
<code>WindDir</code>	Wind direction	$^\circ$	Instantaneous
<code>AirTemp_C_Avg</code>	Air temperature	$^\circ\text{C}$	15-min mean
<code>RH_Avg</code>	Relative humidity	%	15-min mean
<code>Rain_mm_Tot</code>	Rainfall	mm	15-min total
<code>SlrW_Avg</code>	Solar radiation	W m^{-2}	15-min mean
<code>BP_mbar_Avg</code>	Barometric pressure	mbar	15-min mean

3.2 Processing and Analysis

All processing was carried out in Python 3.10+ using `pandas` for data handling, `NumPy` for numerical computation, and `Matplotlib` for visualisation. The workflow proceeds as follows:

- **Load and clean:** the CSV is read with `header=2` to skip the Campbell logger metadata rows. Non-numeric entries (e.g. NAN wind direction readings) are coerced to `NaN` and excluded from aggregation.
- **Daily statistics:** 15-minute records are grouped by date. The mean is taken for temperature, humidity, pressure, wind speed, wind direction and solar radiation; daily total is used for rainfall.
- **Specific humidity (Q5):** derived via the Tetens approximation for saturation vapour pressure and the standard mixing-ratio relation: $q = 0.622 e / (P - 0.378 e)$, where $e = (\text{RH}/100) \times e_s$.
- **Upper-level pressure (Q8):** estimated using the barometric formula $P(z) = P_0 \exp(-z/H)$, with scale height $H = 8400$ m and target altitude $z = 10\,000$ m.

4. Results

4.1 Daily Time-Series (Question 1)

Figure 1 shows the daily mean temperature for the full analysis period. March opens at 24–28 °C and rises steadily through April. The pre-monsoon maximum is reached in May–early June, with several days exceeding 32 °C and peak daily means near 34 °C. Monsoon onset in mid-June suppresses daytime heating; temperatures stabilise at 28–30 °C through July and August with only 1–2 °C transient fluctuations. A gradual post-monsoon cooling trend reduces values to 24–25 °C by late October.

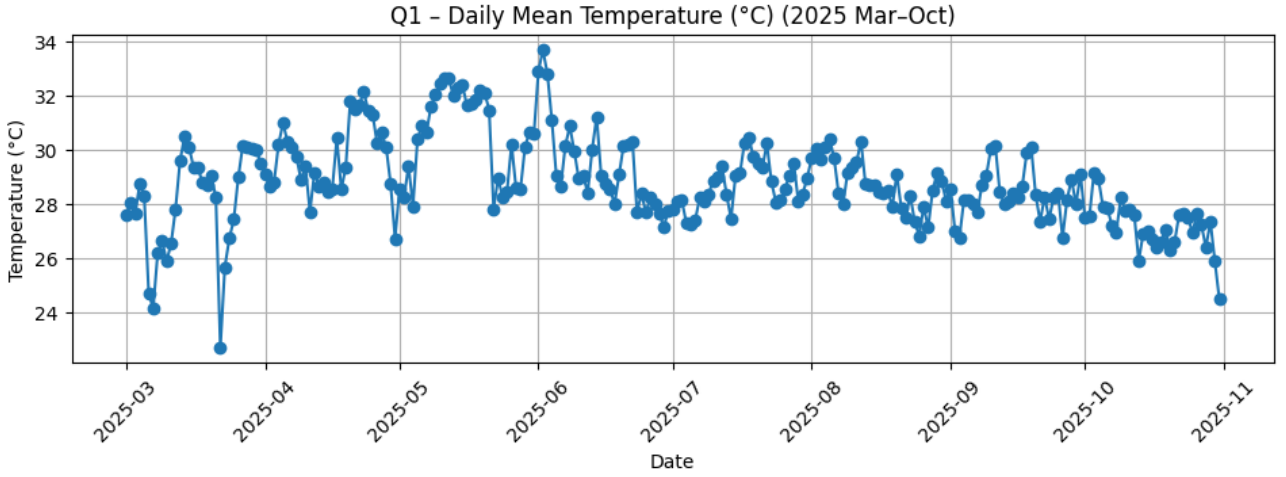


Figure 1: Daily mean air temperature (°C), March–October 2025.

Relative humidity (Figure 2) starts at 60–75% in early March with occasional sharp dips to ~48% attributable to dry continental air intrusions. Humidity increases steadily through April (70–80%) before jumping sharply in late May as monsoon moisture arrives; from June onward daily means frequently reach 88–98%, with values remaining above 90% well into October, suggesting delayed monsoon withdrawal.

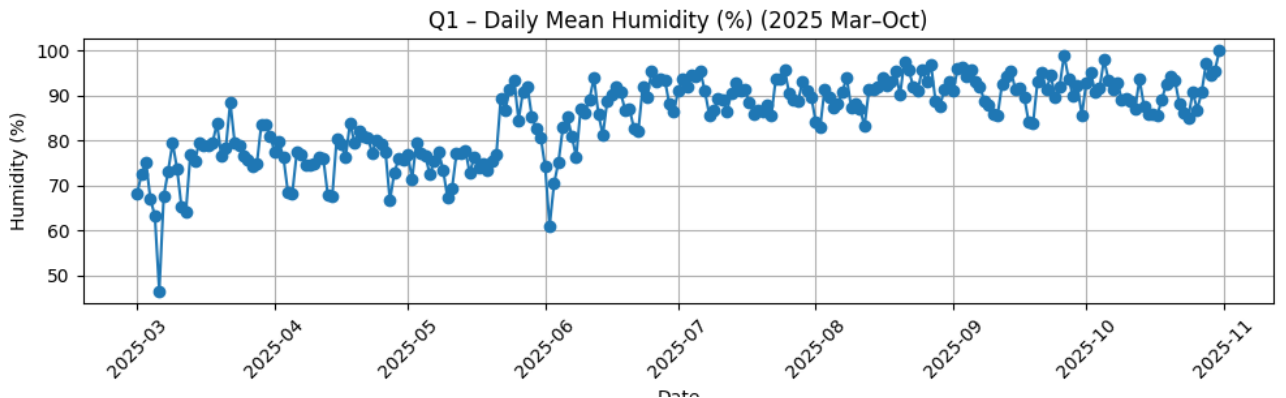


Figure 2: Daily mean relative humidity (%), March–October 2025.

Atmospheric pressure (Figure 3) ranges 994–1000 mbar in March and declines progressively through spring, reaching pronounced minima of 979–981 mbar in late May–July coinciding with the monsoon trough. A slow recovery to 994–996 mbar occurs through October.

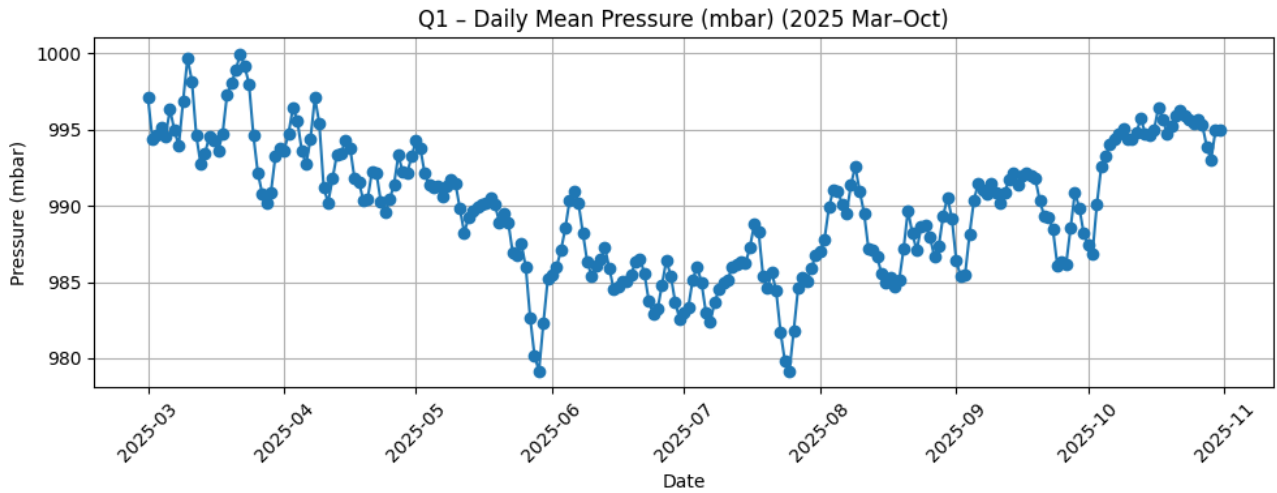


Figure 3: Daily mean atmospheric pressure (mbar), March–October 2025.

Wind speed (Figure 4) peaks in late April–May ($>3 \text{ ms}^{-1}$, isolated maxima $\sim 3.6 \text{ ms}^{-1}$) driven by pre-monsoon convective gusts, then drops sharply to $0.8\text{--}1.5 \text{ ms}^{-1}$ during the monsoon and further to $0.5\text{--}1.0 \text{ ms}^{-1}$ in October.

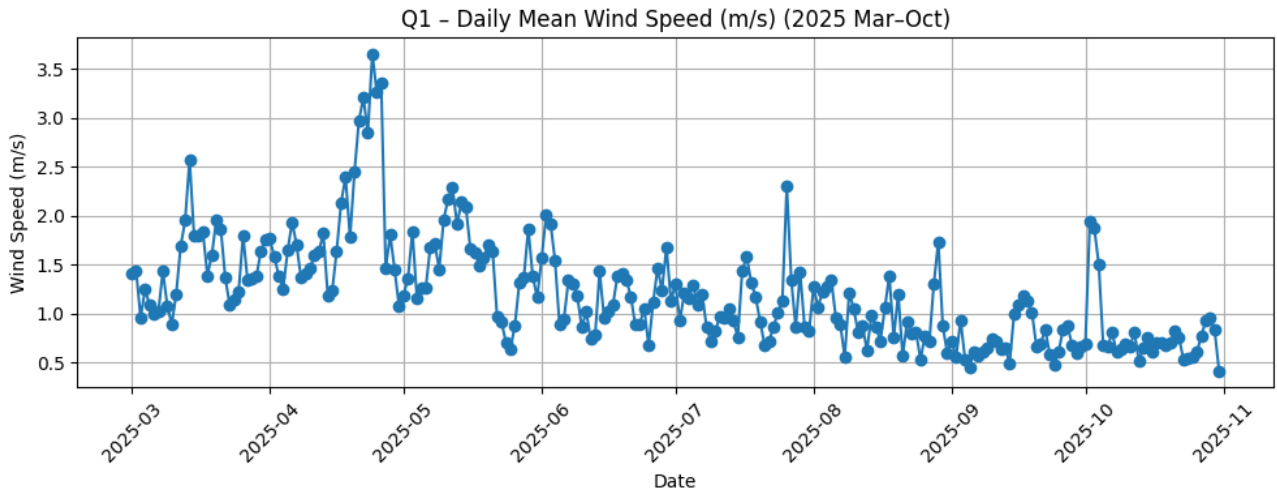


Figure 4: Daily mean wind speed (ms^{-1}), March–October 2025.

Wind direction (Figure 5) clusters between $150^\circ\text{--}210^\circ$ (southeasterly to southwesterly) in March–April, shifts to a broader $110^\circ\text{--}250^\circ$ spread during monsoon transition (May–June), then stabilises at $170^\circ\text{--}220^\circ$ through the active monsoon (July–September) before becoming more variable again in October.

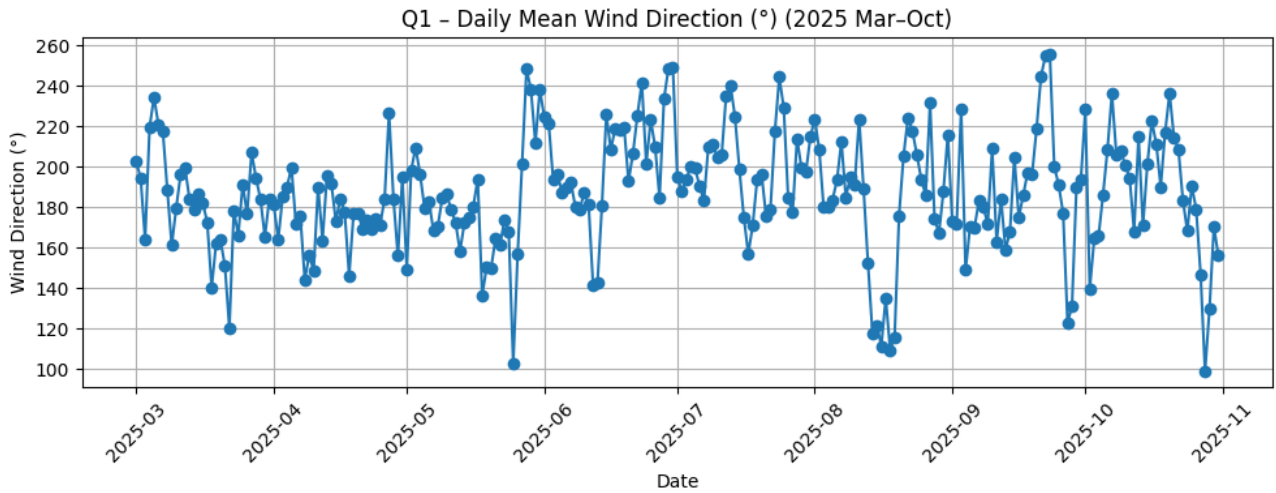


Figure 5: Daily mean wind direction ($^{\circ}$), March–October 2025.

Solar radiation (Figure 6) averages $220\text{--}290\text{ W m}^{-2}$ in March–May under mostly clear skies. The onset of monsoon cloud cover drives an abrupt decline from June onward, where daily means oscillate between $60\text{--}250\text{ W m}^{-2}$ depending on cloud cover. October shows partial recovery to $180\text{--}230\text{ W m}^{-2}$.

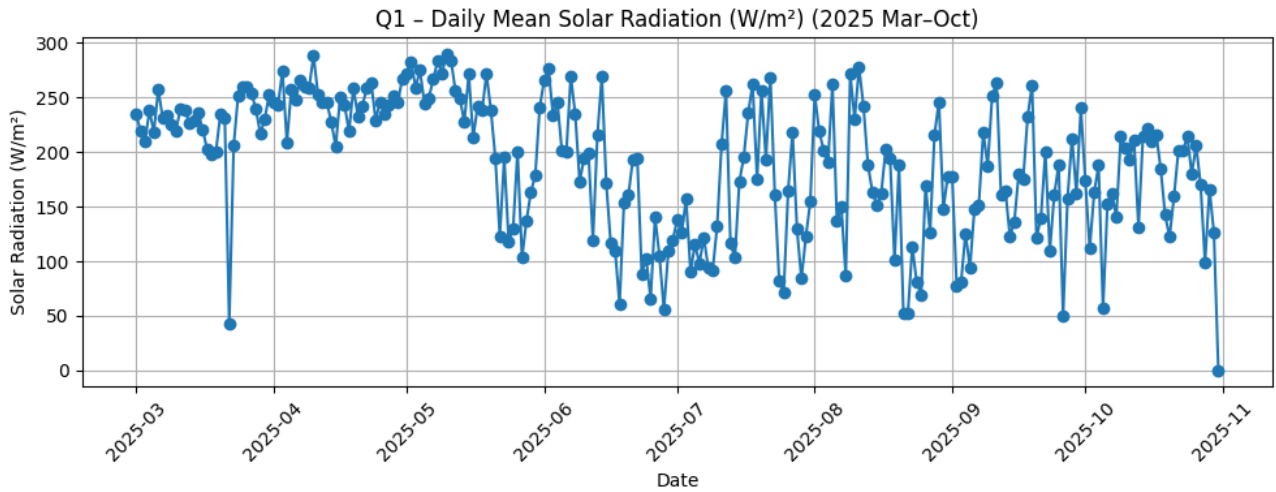


Figure 6: Daily mean solar radiation (W m^{-2}), March–October 2025.

Daily rainfall (Figure 7) is infrequent and light ($<30\text{ mm}$) in March–April, intensifies through May with a sharp onset event around early June ($\sim 55\text{--}60\text{ mm}$), and then shows repeated $10\text{--}40\text{ mm}$ events through September. October retains moderate post-monsoon rainfall before tailing off.

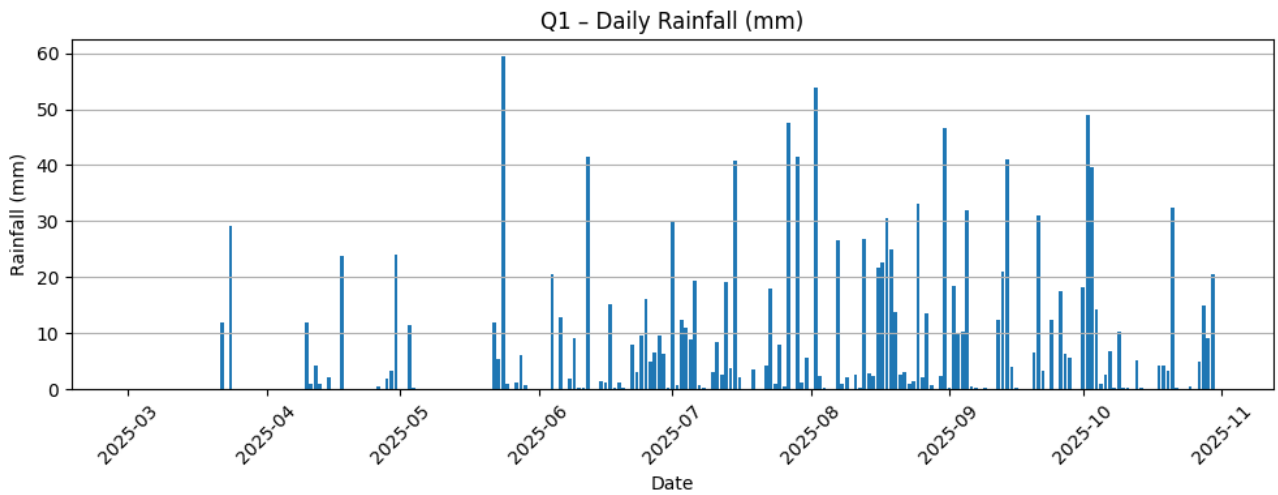


Figure 7: Daily total rainfall (mm), March–October 2025.

Figure 8 overlays solar radiation and daily rainfall on twin axes, illustrating the inverse relationship between insolation and monsoon precipitation.

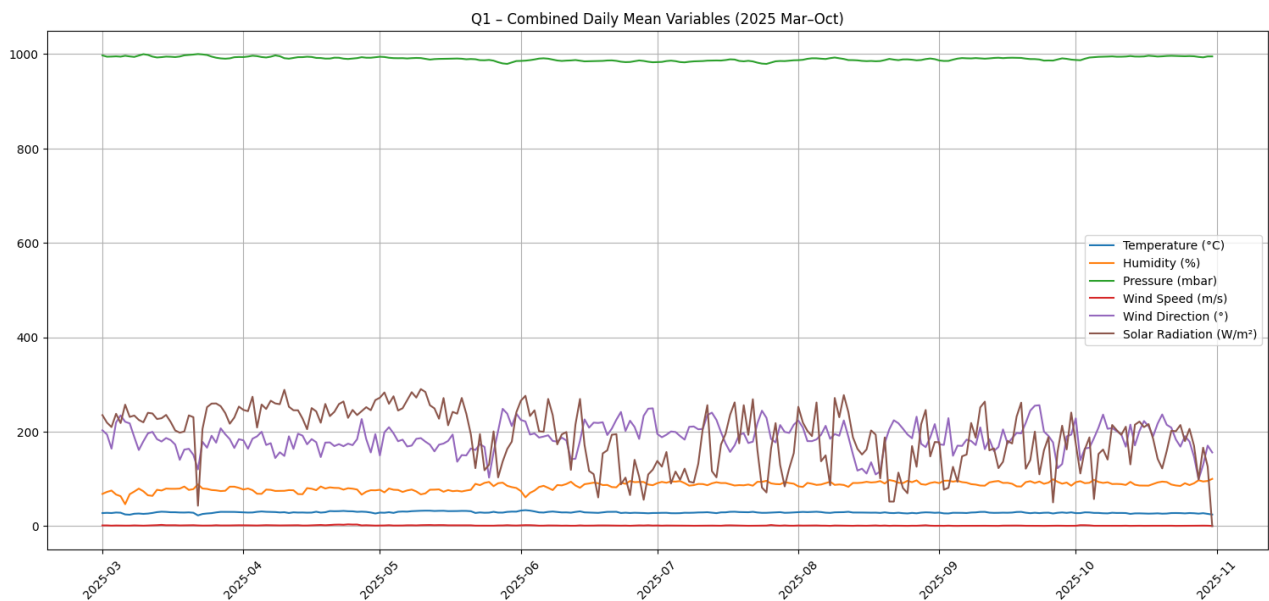


Figure 8: Solar radiation (W m^{-2}) vs. daily rainfall (mm), March–October 2025.

4.2 Global Extremes (Question 2)

Table 2 lists the global maximum and minimum values for each variable and their date–time of occurrence over the full analysis period.

Table 2: Global maximum and minimum values and their occurrence times.

Variable	Maximum	Date / time	Minimum	Date / time
Temperature	41.1 °C	28 Mar 2025, 13:45	15.21 °C	06 Mar 2025, 05:45
Humidity	100 %	04 Mar 2025, 05:30	12.26 %	06 Mar 2025, 12:00
Pressure	1003 mbar	22 Mar 2025, 09:45	977 mbar	29 May 2025, 15:00
Wind speed	7.428 m s ⁻¹	09 Jun 2025, 14:45	0.055 m s ⁻¹	08 Sep 2025, 22:15
Wind direction	359°	18 Mar 2025, 09:15	0°	01 Mar 2025, 22:30
Solar radiation	1069 W m ⁻²	18 Jul 2025, 12:30	0 W m ⁻²	12 Apr 2025, 20:15
Rainfall	37.85 mm	24 May 2025, 12:45	0 mm	01 Mar 2025, 00:00

The highest temperature (41.1 °C) is recorded in late March — notably before the climatological peak month — suggesting an early, intense heat event. The lowest temperature (15.21 °C) falls in the early morning of 6 March, consistent with strong radiative cooling under clear skies. The wind speed maximum (7.428 m s⁻¹) on 9 June aligns with convective activity near monsoon onset.

4.3 Monthly Extremes (Question 3)

Figures 9–14 display monthly maximum and minimum values for each parameter. Key findings are summarised below.

- **Temperature** (Fig. 9): Monthly maxima peak in May–June (~ 40 – 41 °C) and decline to ~ 34 °C by October. Monthly minima rise from ~ 15 °C in March to ~ 25 °C in July, then fall to ~ 21 °C in October.
- **Humidity** (Fig. 10): Monthly maxima remain at $\sim 100\%$ in every month. Monthly minima show a strong seasonal rise from $\sim 12\%$ in March to $\sim 60\%$ in July, falling to $\sim 53\%$ in October.
- **Pressure** (Fig. 11): Maximum pressure drops from ~ 1003 – 1004 mbar in March to ~ 991 – 992 mbar in July, then recovers. Minimum pressure reaches its nadir (~ 976 – 977 mbar) in May–July.
- **Wind speed** (Fig. 12): Maximum wind speeds peak in June (~ 7.5 m s⁻¹) and decline sharply to < 4 m s⁻¹ from July onward.
- **Solar radiation** (Fig. 13): Monthly maxima increase from ~ 940 W m⁻² in March to a peak of ~ 1080 W m⁻² in July, then decline. Monthly minima are near zero in all months, reflecting persistent cloudy episodes.
- **Rainfall** (Fig. 14): Maximum single-day rainfall peaks at ~ 38 mm in May (convective onset storms). Every month contains at least one dry day (minimum = 0 mm).

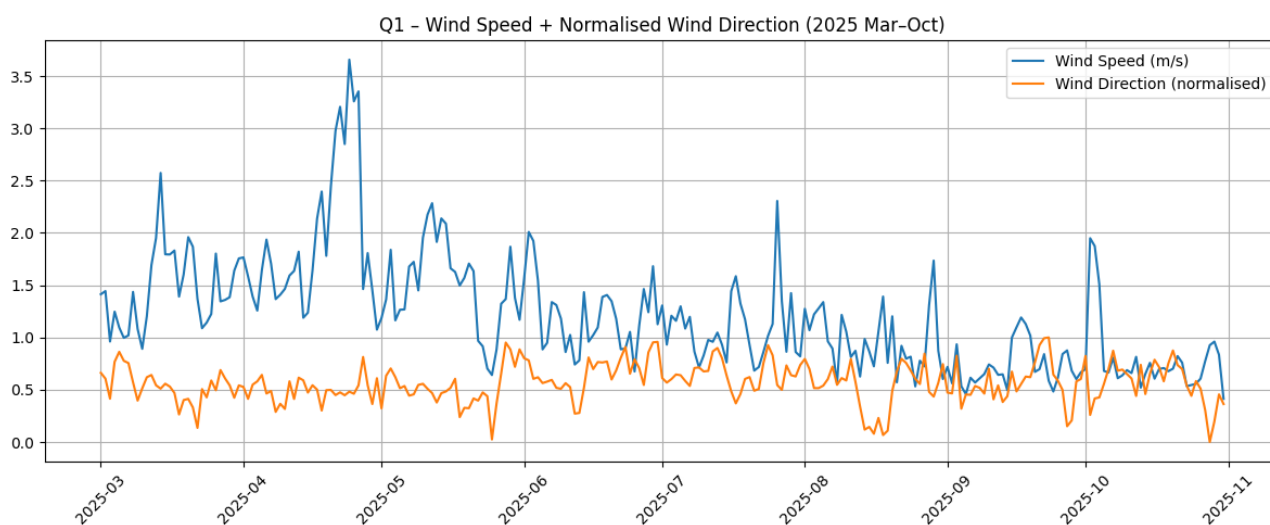


Figure 9: Monthly maximum and minimum temperature ($^{\circ}\text{C}$), 2025.

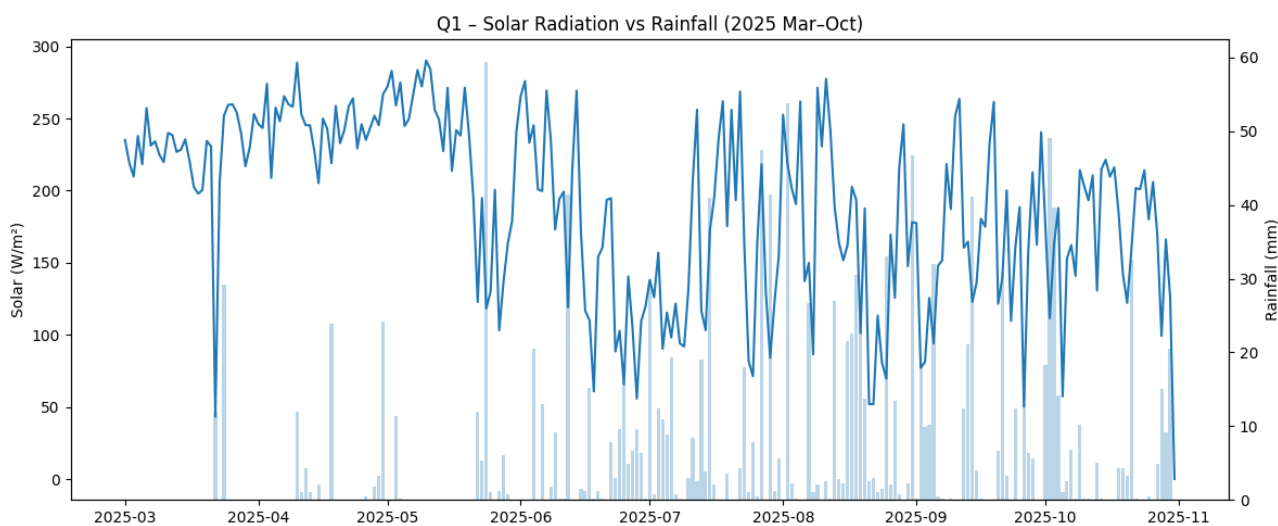


Figure 10: Monthly maximum and minimum relative humidity (%), 2025.

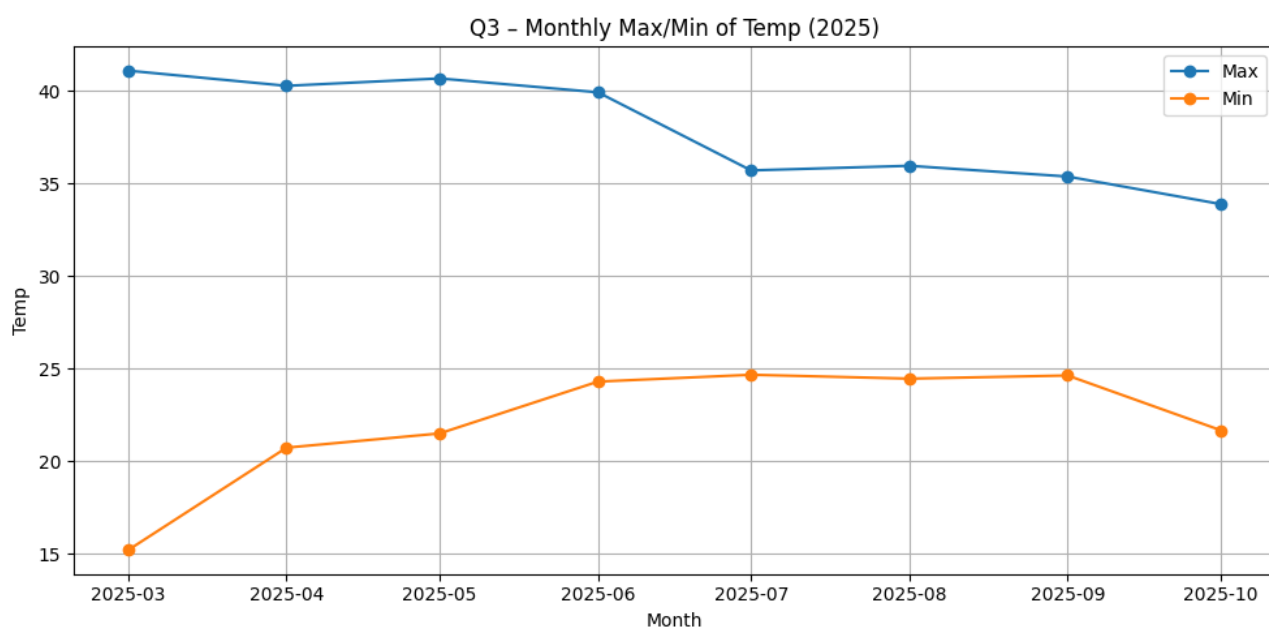


Figure 11: Monthly maximum and minimum atmospheric pressure (mbar), 2025.

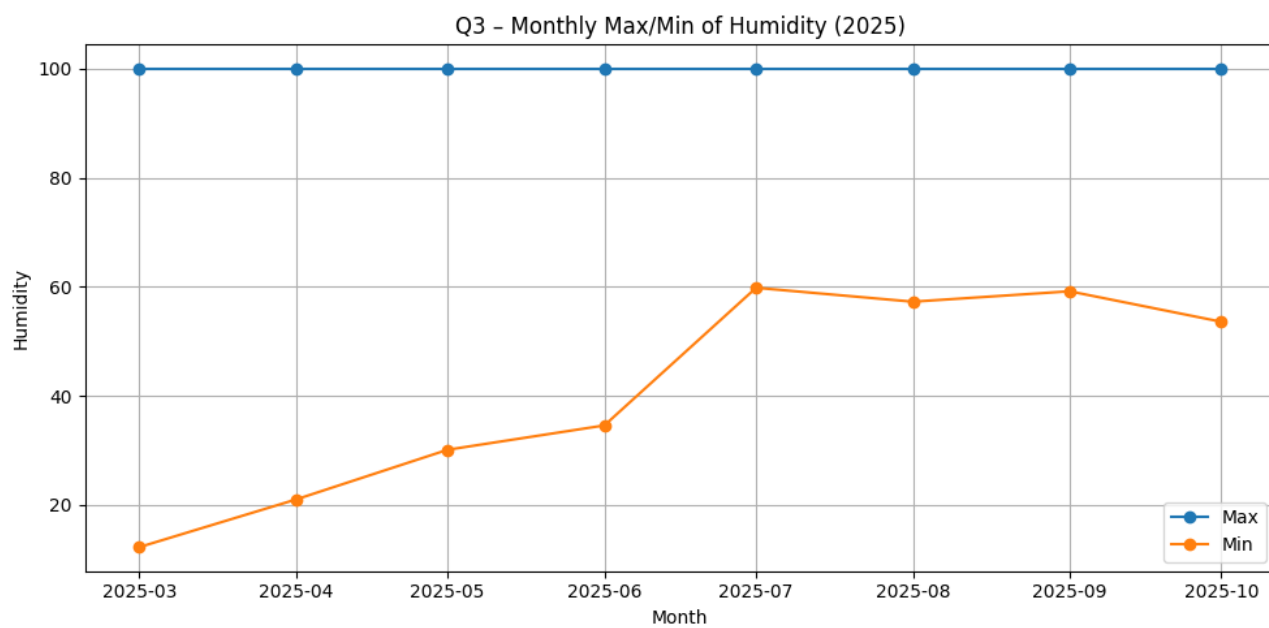


Figure 12: Monthly maximum and minimum wind speed (m s^{-1}), 2025.

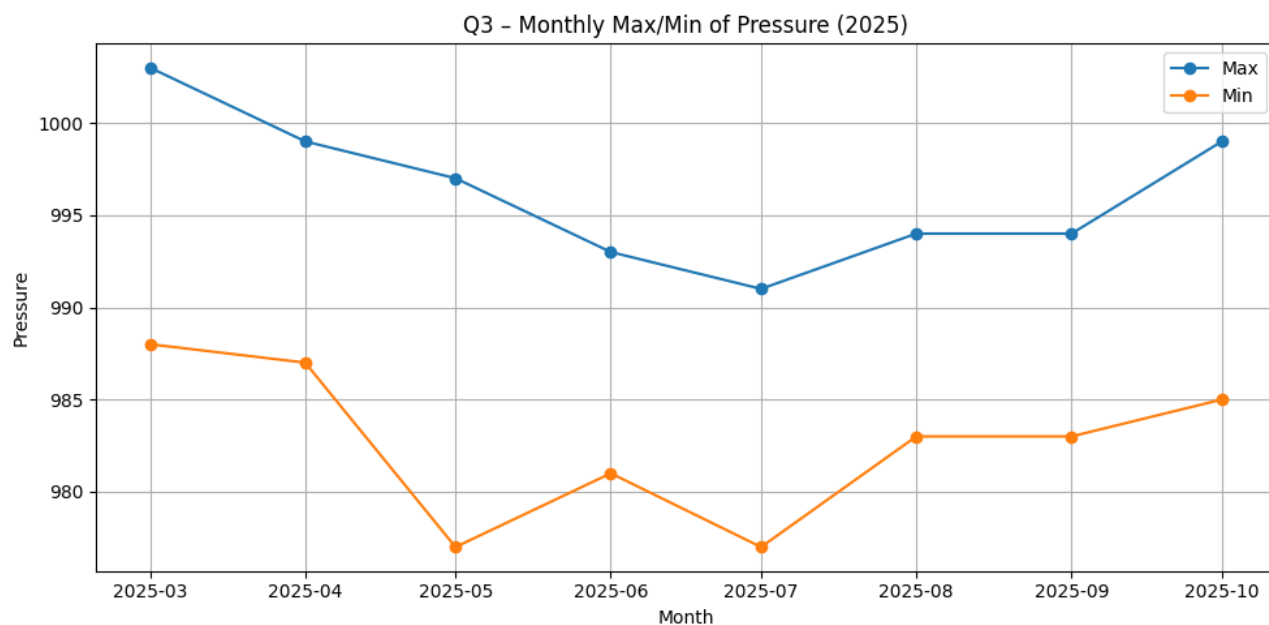


Figure 13: Monthly maximum and minimum solar radiation (W m^{-2}), 2025.

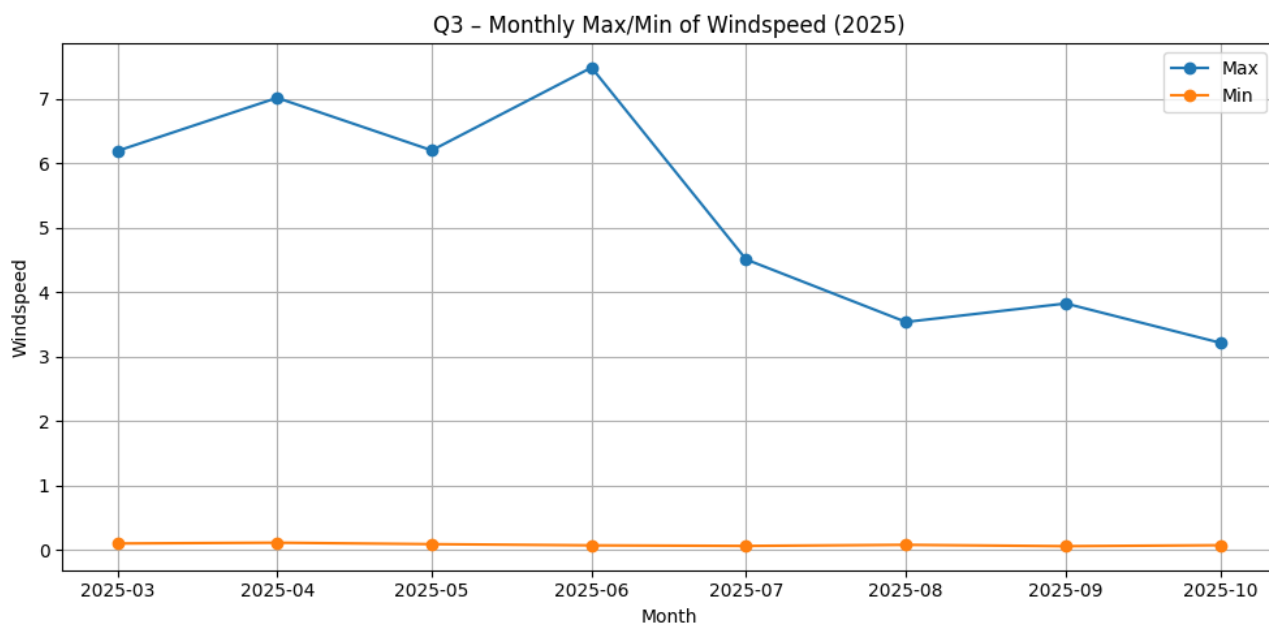


Figure 14: Monthly maximum and minimum daily rainfall (mm), 2025.

4.4 Month of Maximum Rainfall (Question 4)

Aggregating 15-minute rainfall totals by month, **August 2025** receives the highest cumulative rainfall over the analysis period. This is consistent with the climatological peak of the Bay of Bengal monsoon branch over coastal Odisha, which typically delivers its heaviest rainfall in July–August.

4.5 Specific Humidity: Rainy vs. Winter Month (Question 5)

Specific humidity q was computed at every 15-minute time step using the Tetens saturation vapour pressure formula and the standard relation:

$$e_s = 6.112 \exp\left(\frac{17.67 T}{T + 243.5}\right) \quad (1)$$

$$e = \frac{\text{RH}}{100} e_s \quad (2)$$

$$q = \frac{0.622 e}{P - 0.378 e} \quad (3)$$

Figure 15 shows daily mean specific humidity for August 2025 (rainy month) and Figure 16 for October 2025 (post-monsoon reference month).

In August, q is persistently high (0.022–0.0245 kg kg^{−1}) with small day-to-day variability, reflecting the continuously saturated monsoon boundary layer. In October, values are lower (0.018–0.0225 kg kg^{−1}) and significantly more variable, with a minimum near mid-month (~ 0.018 kg kg^{−1}) attributable to intrusions of drier continental air as the monsoon retreats.

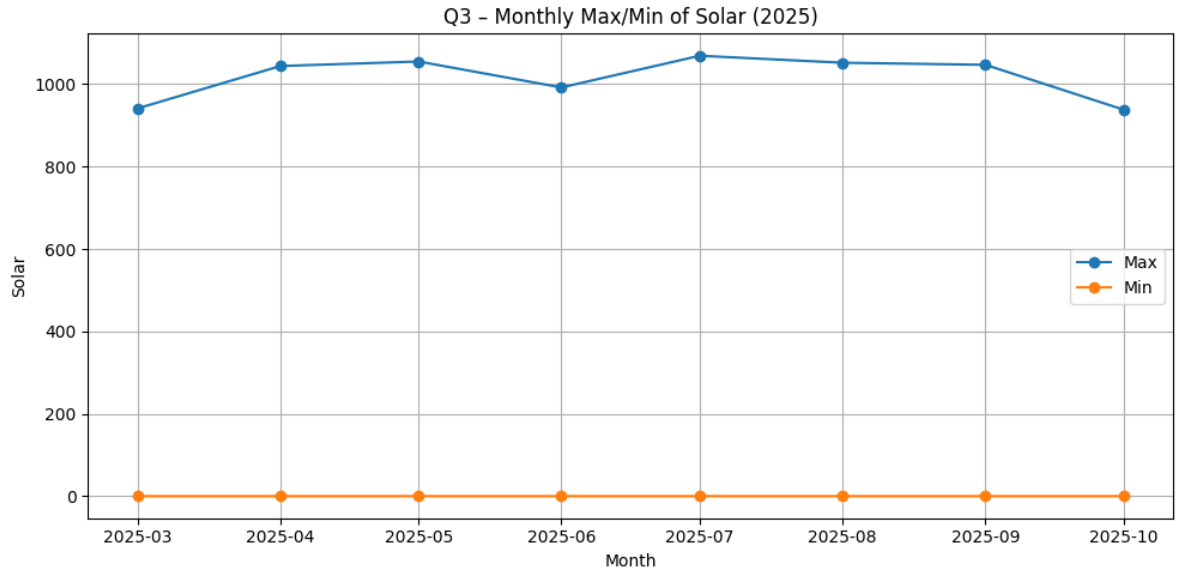


Figure 15: Daily mean specific humidity (kg kg^{−1}) — August 2025 (rainy month).

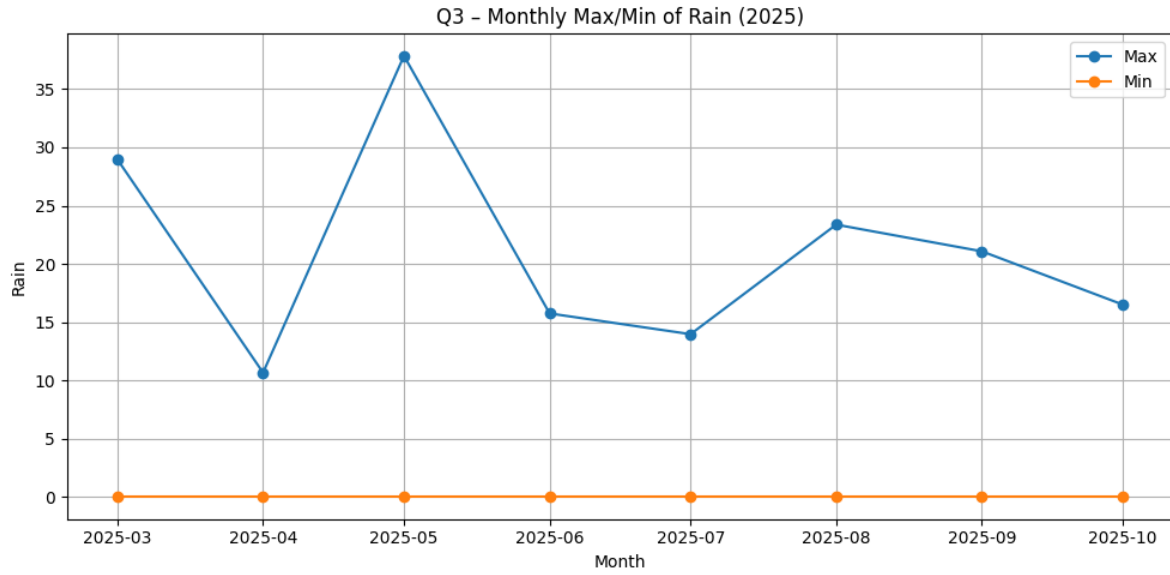


Figure 16: Daily mean specific humidity (kg kg^{-1}) — October 2025 (post-monsoon month).

4.6 Diurnal Variation — August 2025 (Question 6)

Figures 17–21 show the mean diurnal cycle of each variable computed by averaging all 15-minute values at each clock hour across August 2025. Key features are:

- **Temperature** (Fig. 17): minimum at $\sim 05:00$ (26.6°C), rapid morning rise, afternoon peak at $13:00\text{--}14:00$ ($\sim 31.5^\circ\text{C}$), slow evening decay. Night-day range $\approx 5^\circ\text{C}$, muted by monsoon cloud cover.
- **Humidity** (Fig. 18): inversely tracks temperature; nighttime values approach 97–98%, afternoon minimum $\sim 79\%$ at $12:00\text{--}13:00$.
- **Solar radiation** (Fig. 19): classic bell curve; peak at $\sim 11:00$ (580 W m^{-2}), slightly earlier than solar noon due to convective cloud build-up.
- **Pressure** (Fig. 20): classic semi-diurnal atmospheric tide — two maxima ($\sim 09:30$ and $\sim 21:30$) and two minima ($\sim 03:30$ and $\sim 15:30$), amplitude $\sim 2 \text{ mbar}$.
- **Wind speed** (Fig. 21): minimum at dawn ($\sim 05:00$, 0.46 m s^{-1}), peak at $14:00\text{--}15:00$ ($\sim 1.45 \text{ m s}^{-1}$) driven by daytime boundary-layer growth.

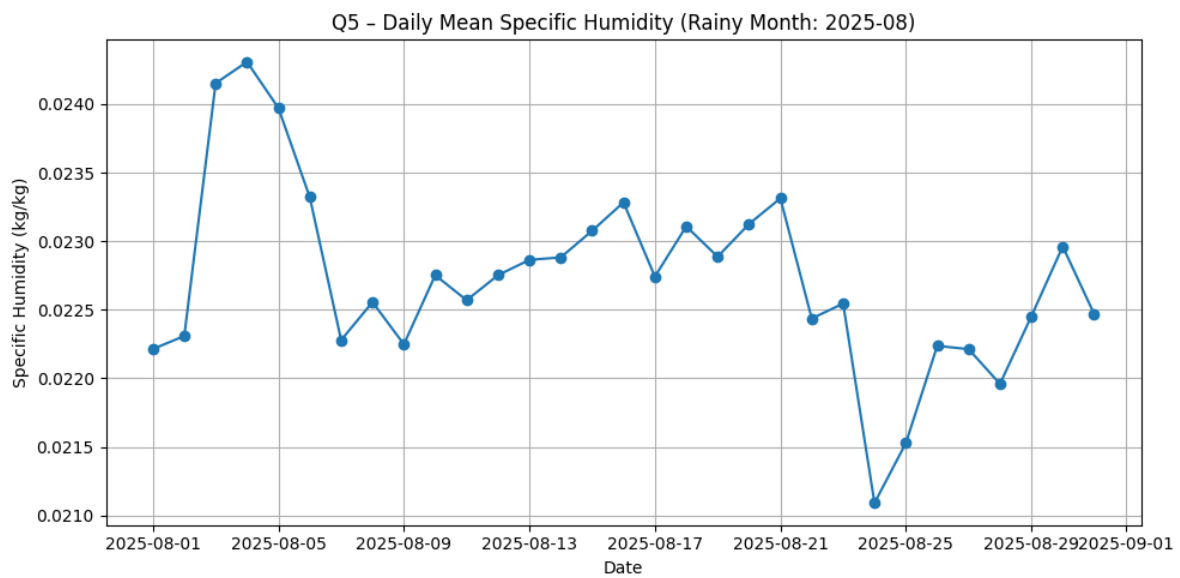


Figure 17: Diurnal mean temperature ($^{\circ}\text{C}$) — August 2025.

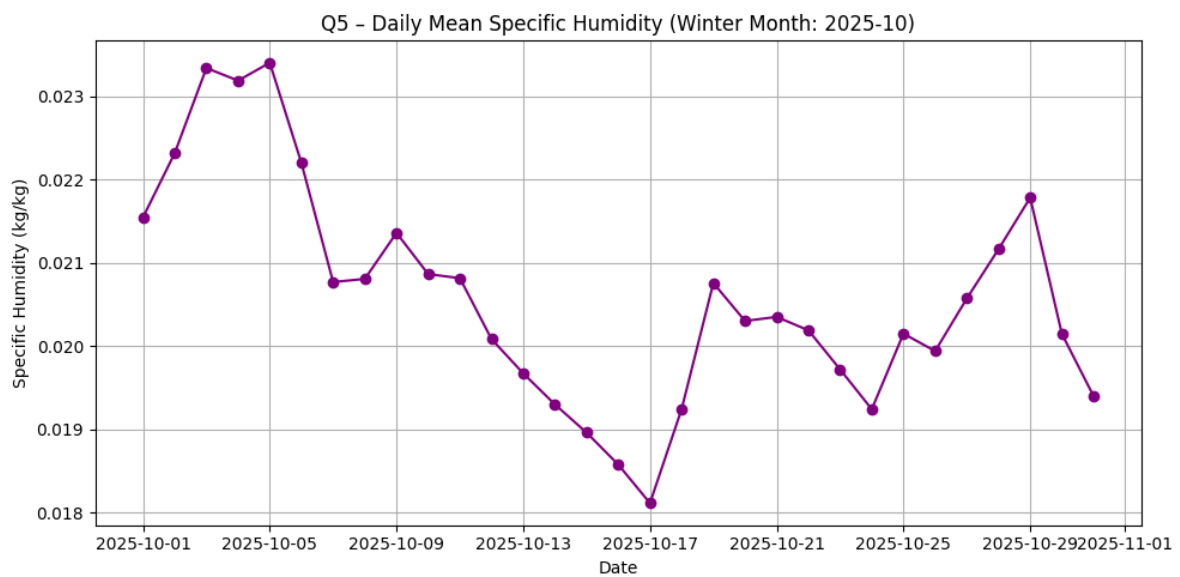


Figure 18: Diurnal mean relative humidity (%) — August 2025.

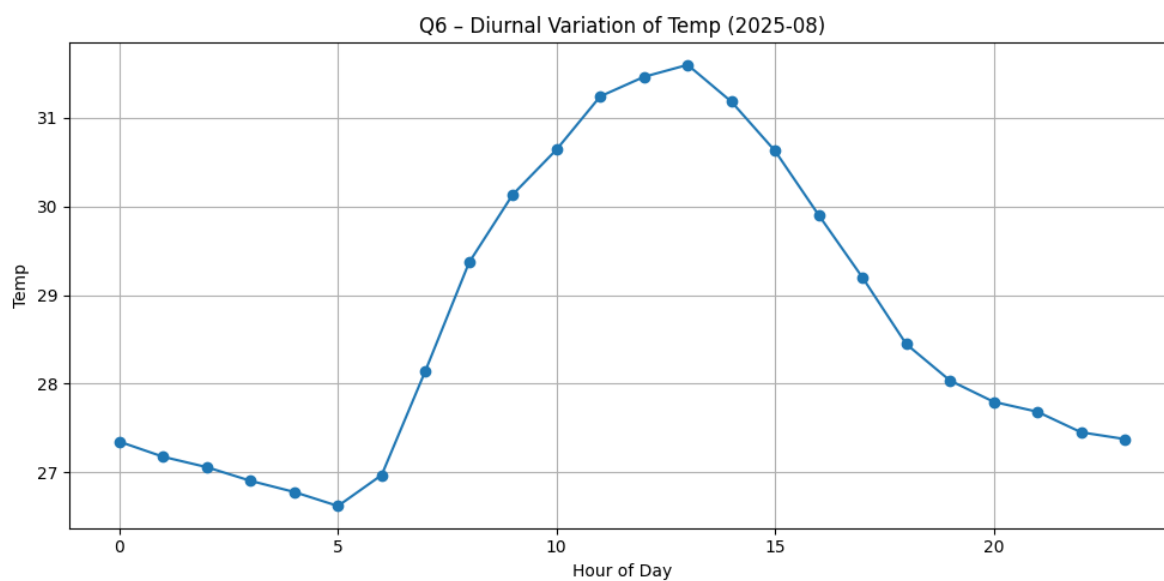


Figure 19: Diurnal mean solar radiation (W m^{-2}) — August 2025.

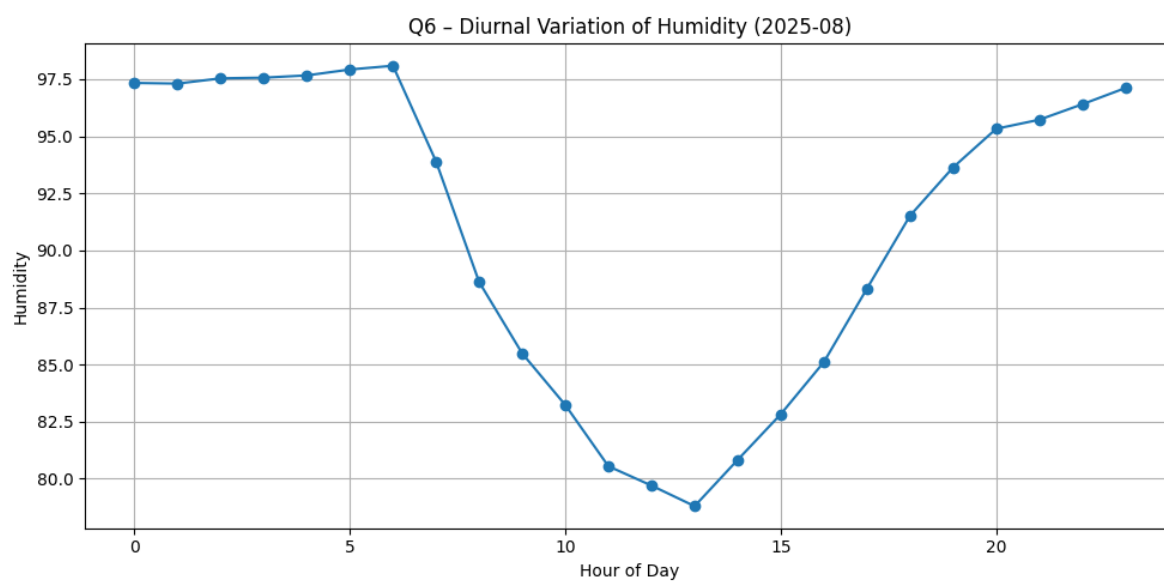


Figure 20: Diurnal mean atmospheric pressure (mbar) — August 2025.

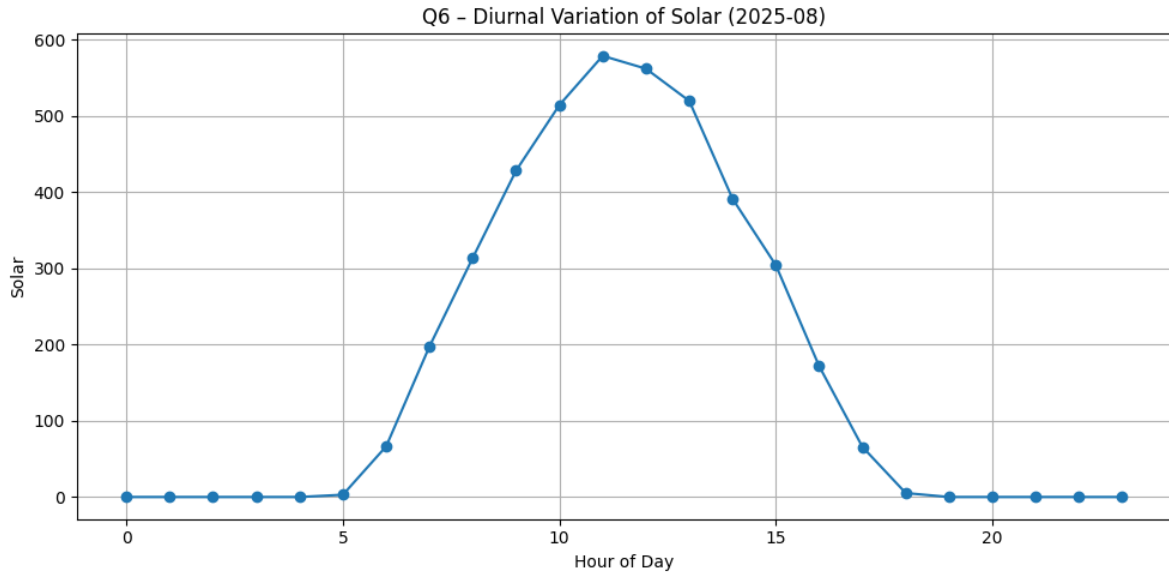


Figure 21: Diurnal mean wind speed (m s^{-1}) — August 2025.

4.7 Cyclone Signature Search (Question 7)

A tropical cyclone at the surface is characterised by a pressure drop of 10–25 mbar within 24 hours, sustained high winds with organised directional rotation, and multi-day continuous heavy rainfall. None of these signatures is found in the dataset.

Two episodes of elevated interest occur: a gradual pressure decline of 4–6 mbar in late May, and a similar shallow trough in late July. In both cases wind speeds remain low ($1\text{--}3 \text{ m s}^{-1}$), rainfall is episodic rather than sustained, and wind direction shows no organised rotation. These features are consistent with the monsoon low-pressure systems and depressions common over the Bay of Bengal in this season.

IMD bulletins for late July 2025 confirm that a cyclonic circulation associated with the remnants of Tropical Cyclone Wipha evolved into a low-pressure system over the north Bay of Bengal, issuing heavy-rainfall warnings for Odisha from 24–27 July. The observational signature at NISER matches a remote monsoon depression rather than a direct cyclone landfall. Figure 22 confirms the absence of cyclone-class signatures.

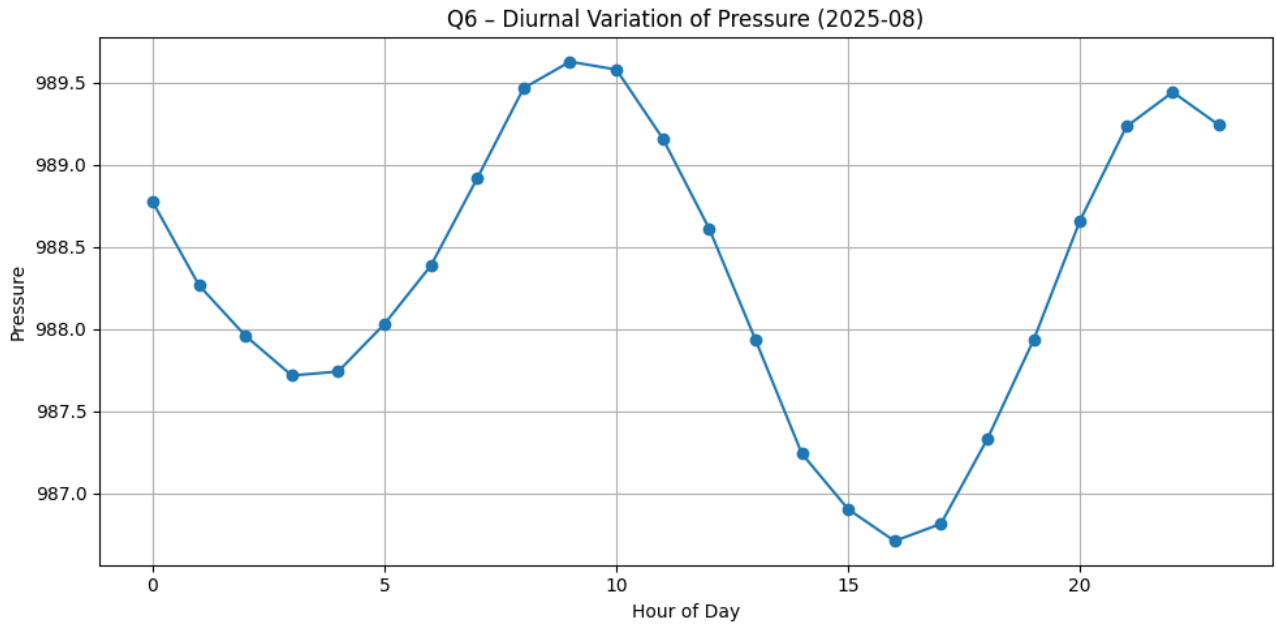


Figure 22: Pressure time-series during the late July 2025 low-pressure episode (Q7).

4.8 Pressure at 10 km Altitude (Question 8)

Using the isothermal barometric formula with a fixed scale height $H = 8400$ m, the surface pressure at each 15-minute interval on 15 July 2025 was extrapolated to $z = 10\,000$ m:

$$P(10\text{ km}) = P_0 \times \exp\left(\frac{-10000}{8400}\right) \approx 0.301 P_0 \quad (4)$$

The resulting curve (Figure 23) oscillates between ~ 299.2 and ~ 300.8 mbar, consistent with upper-tropospheric values. A midday pressure dip corresponds to strong convective outflow raising mass aloft; recovery toward evening reflects subsidence as convection weakens. The ~ 2 mbar diurnal amplitude demonstrates that boundary-layer dynamics are vertically coupled to the upper troposphere even at 10 km.

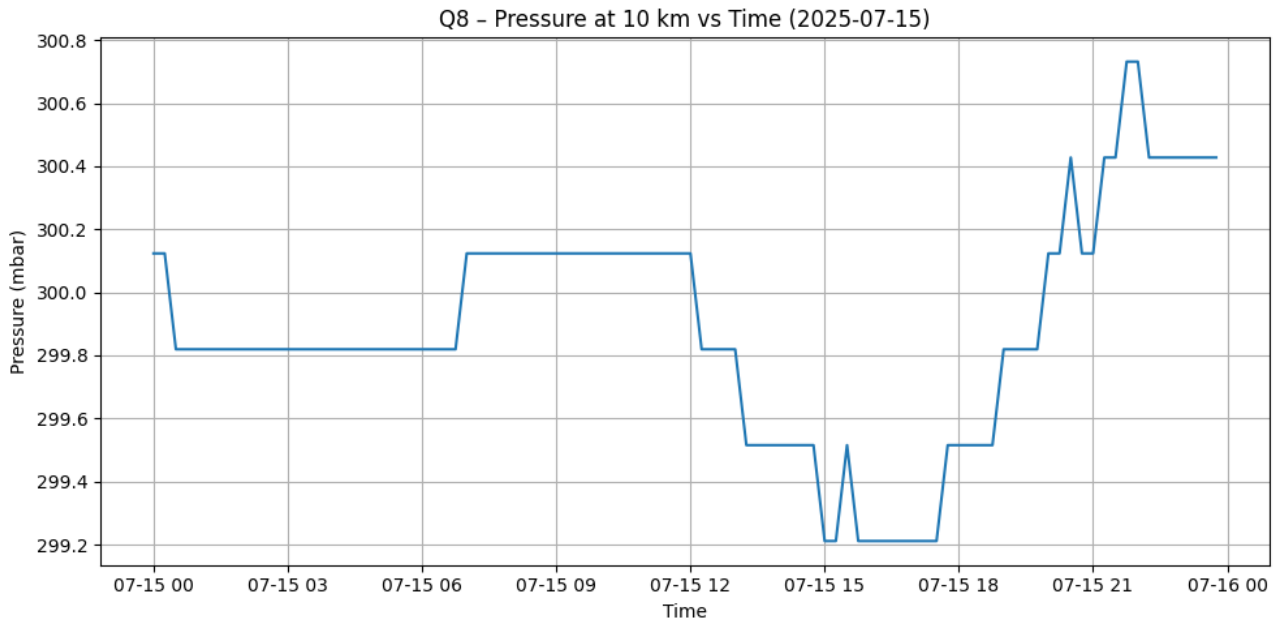


Figure 23: Estimated pressure at 10 km altitude on 15 July 2025, derived from surface pressure via the barometric formula (Eq. 4).

5. Discussion

The eight-month record at NISER Jatni faithfully captures the canonical monsoon cycle of coastal Odisha. Pre-monsoon land heating drives the thermal maximum (41.1 °C in late March) and peak wind activity in May–June. Monsoon onset — identifiable in all variables simultaneously around mid-June — brings suppressed temperature, elevated and stable humidity (88–98%), reduced solar radiation and low surface winds, consistent with the establishment of a moist, well-mixed boundary layer.

The specific humidity analysis confirms that August air holds approximately 15–25% more water vapour per kilogram than October air, quantifying the moisture contrast between peak monsoon and early post-monsoon conditions. The semi-diurnal pressure oscillation apparent in the diurnal analysis (Section 3.6) is a global atmospheric tide feature; its small amplitude (~ 2 mbar) in August is consistent with suppression by monsoon convective activity.

The absence of a cyclone signature in the dataset is noteworthy but unsurprising: Bhubaneswar’s primary cyclone risk window is October–November (post-monsoon), outside the main analysis period. The July low-pressure episode linked to Wipha’s remnants produced heavy rainfall warnings but manifested only as a weak depression at the station level.

6. Conclusions

This study demonstrates the power of high-frequency surface observations for characterising monsoon dynamics at a single station. The key conclusions are:

1. Temperature, pressure and wind speed show coherent seasonal cycles driven by pre-monsoon heating and monsoon circulation, with onset and retreat clearly identifiable in the time series.
2. August 2025 received the highest cumulative rainfall; specific humidity during this month ($0.022\text{--}0.0245\text{ kg kg}^{-1}$) was 15–25% higher than in October, the selected post-monsoon reference month.
3. Diurnal cycles in August follow textbook boundary-layer behaviour, but with amplitudes modu-

lated by persistent cloud cover and moisture.

4. No tropical-cyclone-class pressure or wind signature was detected. Two monsoon low-pressure episodes are consistent with typical Bay of Bengal depressions rather than cyclone landfall.
5. Upper-tropospheric pressure at 10 km, estimated via the barometric formula, exhibits a ~ 2 mbar diurnal oscillation on 15 July 2025, reflecting vertical coupling of boundary-layer convection to the upper troposphere.

7. Acknowledgements

The author thanks Dr. Jayesh Mahendra Goyal for course guidance and the NISER infrastructure team for maintaining the AWS and making the data available.